ECON500 EXPOSITORY PAPER

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Abstract. In this expository article, we offer a stream-lined view of the classic principal agent problem, starting with the canonical two-by-two case with 2 effort levels from the agent (high and low), moving on to discrete time cases, multiple interaction, and to continuous time. In the latter half of the article, we introduce more recent arguments using tools such as stochastic calculus and dynamic programming to analyze the principal agent problem with hidden action. We close this narrative by presenting a closed form solution to a special case of Sannikov's stochastic principal-agent model, and we offer interpretations and discussions relating to the rising rising wage profile arguments.

1. Introduction and Motivation

The principal agent problem is a classic problem that has many applications. It essentially describes how a "principal" (which could be thought of as shareholders) could optimally hire an "agent" (which could be thought of as a manager) so that the principal could maximize her utility while share risk with the agent. We add a layer of complexity to the problem by introducing hidden action and hidden type. The problem is also called *moral hazard* when the agent has private action, and the case of hidden type is also known as *adverse selection*.

A benchmark one period model of the principal agent problem could be found in textbooks such as Bolton and Dewatripont (2004). Since a principal could choose to renew the contract at the end of each period, say, two years, this problem could be readily generalized to a multi-period model, as in Rogerson (1985) and Spear and Srivastava (1987). In general, the optimal multi-period contract could be much more complex than the single period one. This complexity is contrasted with the simplicity of long-term incentive contracts observed in reality. Long term employment contracts observed in practice do not appear

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to fine-tune the agent's intertemporal consumption to the extent predicted by the model. Thus, it is good news that this problem is not only studied in the theoretical setting, but also studied in many applications. For example, in EDMANS et al. (2012), the authors used descrete time models and applied contract theory to CEO compensation, where under certain assumptions, they obtained a closed-form contract that yields clear predictions for how "level and performance sensitivity of pay vary over time and across firms."

We illustrated that the principal agent problem can be naturally framed in a discrete time model, since in the real world, the principal signs a contract which often lasts one year or more with the agent. Hence, we could consider repeated interaction between the principal and the agent period by period. As the frequency of compensation, contracting, and measurement of output increases, we could adapt the model to continuous frame work, where time can take a continuum of values. The literature in continuous-time models started with an important paper by Holmstrom and Milgrom (1987), where they assumed the principal and the agent both have exponential utilities, and they proved that the optimal contract is linear. In more recent developments such as Sannikov (2008) and Hu, Ren and Touzi (2019), the authors formulated the problem as a stochastic control problem, where they investigated the problem using stochastic differential equations.

2. Literature Review

2.1. **The Two by Two case.** Let us investigate the most salient moral hazard problem where both effort and possible levels of production are binary. In this simplified setting, we could focus on the key dynamics of this problem. Our main reference for this two by two case is Laffont and Martimort (2002). Let the principal assign a task to the agent. Now, the output of the task takes two possible values, a high value $Q \in \mathbb{R}$ and a low value $q \in \mathbb{R}$, where $Q > q > 0$. The output depends on probability and the agent's effort, which takes value 1 or 0, which is high effort, or no effort. The main charateristic of the problem is that agent's action is not observable for the principal. The output is Q, with probability p_0, if the agent makes 0 effort, and with probability p_1 if the agent makes 1 effort. Also, $p_0 < p_1$. The cost of effort for the agent is c_0 if he makes 0 effort, and the cost of effort is c_1 if he makes 1 effort. Also, $c_0 < c_1$. Hence, we could summarize this data in table (2.1).

	high output Q	low output q
no effort	p_0	$1-p_0$
high effort	p_1	$1-p_1$

	expected payoff
no effort	$p_0 \cdot u(M) + (1-p_0)u(m) - c_0$
high effort	$p_1 \cdot u(M) + (1-p_1)u(m) - c_1$

The principal would like to design a contract that incentivize the agent to choose effort 1. However, the principal could only infer the agent's action from the output level she observes. The principal would like to design a contract that pays the agent M if the output is Q, and pays the agent m if output is q. Let $u(\cdot)$ denote the utility function for the agent, and let us assume $u(\cdot)$ is increasing and concave, which matches the agent being risk averse. Now, we put the expected payoffs for the agent into table (2.1).

In order for the agent to enter the contract, the two level of payoffs need to satisfy the following two equations, where $\bar{U}$ denotes the agent's next best outside option.

$$\text{IC: } p_1u(M) + (1-p_1)u(m) - c_1 \geq p_0u(M) + (1-p_0)u(m), \tag{1}$$

$$\text{IR: } p_1u(M) + (1-p_1)u(m) - c_1 \geq \bar{U}. \tag{2}$$

IC stands for incentive compatibility, and it says the payoff the agent get from high effort needs to be greater equal to the payoff he would get if he puts in 0 effort. IR stands for individual rationality, which says entering the contract and supplying high effort should give the agent a payoff that is at least his next best option, since otherwise, the agent would not enter the contract in the first place.

Assuming the principal is able to induce effort from the agent, then the principal's payoff (value) function V_1 would be

$$V_1 = \max_{m,M} p_1(Q-M) + (1-p_1)(q-m). \tag{3}$$

Hence, the principal has control over setting the compensation m and M that the agent is going to obtain, hence it hopefully induces effort from the agent. Note that we need V_1 to satisfy $V_1 \geq p_0Q + (1-p_0)q$ since otherwise, the principal could set $M = m = 0$ so as to induce zero effort from the agent, effectively "retiring" the agent.

2.1.1. *Complete Information with observable effort.* As a benchmark case, we assume the principal could directly observe the agent's effort. Hence, agent action is verifiable, and could be included into a contract

that is enforced by the court of law. Hence, the principal's problem is to maximize

$$V_1 = \max_{m,M} p_1(Q - M) + (1 - p_1)(q - m), \tag{4}$$

subject to the **IR** constraint, which is also called participation constraints. Now, we first take derivative with respect to M, and let λ denote the Lagrange multiplier, we have

$$-p_1 + \lambda p_1 u'(M) = 0 \tag{5}$$

And we also take derivative with respect to m, and we have

$$(1 - p_1) + \lambda(1 - p_1)u'(m) = 0. \tag{6}$$

Rearranging, we obtain

$$u'(M) = u'(m) = \frac{1}{\lambda}, \tag{7}$$

and this expression is independent of the output level Q, q. Hence, the agent obtains **full insurance** from the principal, meaning that the agent marginal utility the agent obtains from the payment M, m is constant, while the output could vary between Q and q. Hence, the agent is insured against any output shocks. Also, $M = m$ in this benchmark setting. Hence, with verifiable effort, the risk-averse agent gets full insurance from the risk-neutral principal, and the payment m is the same no matter what production outcome realizes.

2.1.2. *Trade Off between Insurance and Efficiency.* Without full information, the agent's risk aversion becomes a source of inefficiency. Let us write $u(M) \equiv U$, $M \equiv u^{-1}(U) \equiv h(U)$, and $u(m) \equiv u$, $m \equiv h(u)$. Those new variables are levels of ex post utility obtained by the agent. Now, we could restate the **IC** constraint as

$$p_1 U + (1 - p_1)u - c_1 \geq p_0 U + (1 - p_0)u, \tag{8}$$

and we could restate the **IR** constraint as

$$p_1 U + (1 - p_1)u - c_1 \geq \bar{U}. \tag{9}$$

Hence, the principal's problem becomes

$$max_{u,U} p_1(Q - h(U)) + (1 - p_1)(q - h(u)), \tag{10}$$

subject to the restated IR and IC constraint above. Since $h(\dot{})$ is strictly convex, the principal's value function is strictly concave in (u, U). The constraints are linear, and the interior of the constrained set is nonempty, so we have a concave problem. Let λ and μ be non-negative

multipliers associated with the IR constraint and the IC constraint, then taking derivative with respect to U, we have

$$-p_1 h'(U) + \lambda(p_1 - p_0) + \mu p_1 = -\frac{p_1}{u'(M)} + \lambda(p_1 - p_0) + \mu p_1 = 0 \tag{11}$$

and taking derivative with respect to u gives

$$-(1-p_1)h'(u) - \lambda(p_1 - p_0) + \mu(1-p_1) = -\frac{(1-p_1)}{u'(m)} - \lambda(p_1 - p_0) + \mu(1-p_1) = 0 \tag{12}$$

Here, m and M are second-best contract payment to the agent. Rearranging and let $\Delta p \equiv p_1 - p_0$, we get

$$\frac{1}{u'(M)} = \mu + \lambda \frac{\Delta p}{p_1}, \tag{13}$$

$$\frac{1}{u'(m)} = \mu - \lambda \frac{\Delta p}{1-p_1}. \tag{14}$$

Solving for (M, m, λ, μ), we obtain

$$\mu = \frac{p_1}{u'(M)} + \frac{1-p_1}{u'(m)} > 0, \tag{15}$$

and

$$\lambda = \frac{p_1(1-p_1)}{\Delta p}\left(\frac{1}{u'(M)} - \frac{1}{u'(m)}\right), \tag{16}$$

and

$$M = h\left(c_1 + (1-p_1)\frac{c_1}{p_1 - p_0}\right) \tag{17}$$

$$m = h\left(c_1 - p_1\frac{c_1}{p_1 - p_0}\right). \tag{18}$$

Here, the risk-averse agent does not receive full insurance anymore. This result must be contrasted with what we see under complete information earlier. In the first-best case, the agent is paid $u^{-1}(c_1) = h(c_1)$ regardless of the output level being Q or q. In the second-best case with inefficiency from risk aversion, the agent is paid more than $h(c_1) = u^{-1}(c_1)$ when the output is high (Q), and paid less when the output is low (q). When IR is binding and by Jensen's inequality, we have

$$c_1 = p_1 u(M) + (1-p_1)u(m) < u\left(p_1 M + (1-p_1)m\right), \tag{19}$$

so the expected payment $p_1 M + (1-p_1)m$ from the principal to the agent is larger than the first-best cost $u^{-1}(c_1) = h(c_1)$, indicating that the principal is paying the agent **a risk premium** compared to the

first-best case, to compensate the risk-averse agent for bearing the risk of getting the smaller payout $m < M$.

2.2. **Moral hazard problem discrete time one period.** We will refer to the principal as "she" and the agent "he". We refer to the textbook Salanié (2005), Bolton and Dewatripont (2004), and Cvitanic and Zhang (2014) for the single period principle-agent (PA) problem and the continous-time version of the problem. The basic set up for the single period problem is as follows: The principal needs to write a contract such that

- the agent will want to accept the contract, this is termed **IR constraint**, which stands for individual rationality constraint, or participation constraint.
- the principal will maximize her expected utility by getting the most out of the agent's performance. This depends on three classes of models: **RS: risk sharing** with symmetric information, **HA: hidden action** and **HT: hidden type**.
- **Risk Sharing** is also called **first best**, where the Principal and the Agent have the same information, which means the Principal can observe the Agent's action. Hence, she can essentially make the agent to choose the efficient action, and choose the wage to achieve optimal risk sharing.
- **Hidden Action** is also called **second best**, where the Principal can only observe the outcome correlated with the Agent's action, but not Agent's action itself. In real life, this could due to the high cost of monitoring the Agent, or the impossible task of deducing how much effort (action) the Agent had put forth. In the case that the action is observed, but cannot be contracted upon, then it is still second best. Hence, the Principal offers a contract to the Agent that trades off risk sharing and incentives.
- **Hidden type** combined with hidden actions is also called **third best**. Hidden type refers to the situation where the Principal does not know the characteristics/types of the Agent. Hence, Principal will offer **a menu of contracts**, one for each type, for the Agent to pick. When the agent pretend to be of a different type than he really is, it is called **adverse selection**.

We first consider a standard model in finite, discrete version from Chapter 5 of Salanié (2005). The Agent choose between n actions: $a_1, \ldots, a_n$. These actions produce outcomes $x_1, \ldots, x_m$. We assume when the Agent choose action a_i, the Principal observes outcome x_j with probability $p_{ij} > 0$. Only outcome is publicly observed, so the

contract takes the form of a wage that only depends on outcome. If the Principal observes the outcome x_j, she will pay the Agent a wage w_j, and keep $(x_j - w_j)$ for herself (outcome minus wage paid to Agent).

Let $u(w, a)$ denote Agent's utility function, and we assume function u is separable in income w and action a. We could take Agent's utility function in the following form:

$$u(w, a) := u(w) - a,$$

where $u(w)$ is increasing and concave. We assume the Principal is risk neutral, and her utility function is

$$x - w,$$

which is outcome minus the wage paid to the agent.

The Principal offers the Agent a contract $\{w_j\}_{j\in[m]}$, then the Agent chooses his action by solving

$$\max_{i\in\{1,\ldots,n\}} \left(\sum_{j=1}^{m} p_{ij} u(w_j) - a_i \right), \tag{20}$$

this equation is saying that if there's no further constraints, then the agent will pick action a_i among $a_1, \ldots, a_n$ such that the expected utility (from wage minus action) is maximized. The **IC** (incentive constraints) takes the form

$$\sum_{j=1}^{m} p_{ij} u(w_j) - a_i \geq \sum_{j=1}^{m} p_{kj} u(w_j) - a_k, \tag{21}$$

where $k \neq i$ and $k \in [n]$. These $n - 1$ equations are essentially the same as the condition in equation 20. We also need the **IR** individual rationality constraint. Let $\underline{U}$ denote the utility the Agent could obtain by using his outside option/ pursuing his next best opportunity. Then **IR** could be written as

$$\sum_{j=1}^{m} p_{ij} u(w_j) - a_i \geq \underline{U}, \tag{22}$$

for the action a_i that the agent picked.

On the other hand, the Principal choose the contract $(w_1, \ldots, w_m)$ that maximizes her expected utility, while taking into account the consequences of this contract on Agent's actions.

$$\max_{w_1,\ldots,w_m,i} \sum_{j=1}^{m} p_{ij}(x_j - w_j), \tag{23}$$

under the constraints **IC** and **IR**. Let λ_k denote the Lagrange multiplier for **IC** for each k, and let μ denote the multiplier for **IR**. We fix a_i, and the Lagrangian for the Principle is

$$\begin{aligned} L(w, \lambda, \mu) = &\sum_{j=1}^{m} p_{ij}(x_j - w_j) \\ &+ \sum_{k \neq i} \lambda_k \left(\sum_j p_{ij} u(w_j) - a_i - \sum_j p_{kj} u(w_j) + a_k \right) \\ &+ \mu \left(\sum_j p_{ij} u(w_j) - a_i - \underline{U} \right). \end{aligned}$$

We get the first order condition by differentiating, and at **first best**, we would get efficient risk-sharing, where the ratio of marginal utilities of the Principal and the Agent would be constant, which implies the wage w is constant.

$$\frac{1}{u'(w_j)} = \mu_0, \tag{24}$$

where μ_0 is chosen so that **IR** constraint is an equality.

2.3. **Moral hazard problem discrete time multiple period.** In reality, many principle-agent relationships would repeat over many periods, and one example would be an executive working for the board of a company for many years. Such repetition changes the underlying incentive problem in three ways.

- Repetition can make the agent less risk averse, since he could offset a bad outcome in one period by borrowing against future income. Hence, there could be intertemporal risk sharing.
- Repeated observations of the output give more information about the agent's action (effort), hence lead to better contracts.
- Repetition increases the agent's set of available actions. Why? Because the agent could choose when to work, and offset any bad outcome in one period by working harder in the next period.

One reference for this multi-period model is Bolton and Dewatripont (2004), and we have Rogerson (1985), and another is Spear and Srivastava (1987). We start with the two-period model. At the beginning of period $t = 1, 2$, the agent privately chooses an action $a_t \in A$. we have n possible outcomes for this action a, and outcome $x_i \in \{x_1, \ldots, x_n\}$ happens with probability $p_i(a_t)$. $t \in \{1, 2\}$ denotes the period, and $i \in \{1, \ldots, n\}$ is index for the outcome. We assume the probabilities $p_i(a) > 0$ for all action $a \in A$. In each period, the agent has objective function

$$u(w) - \psi(a), \tag{25}$$

where $u(w)$ is the utility of the agent from getting wage (contract) w, and $\psi(a)$ is the cost of action (effort). The principal has objective function

$$V(x - w),$$

where x is the outcome, and w is the wage (contract) given to the agent. The timing of the contracts is as follows:

- At the start of period $t = 1$, the principal make a take-it-or-leave-it offer of a long term contract to the agent. The contract specify action a_1 for the first period, and action plan for the section period $a_2(q^{(1)})$. The contract also specify wage $w_1(q^{(1)})$, and $w_2(q^{(1)}, q^{(2)})$. $q^{(1)}, q^{(2)}$ denotes the probability distribution of outcome in period 1 and period 2 respectively.
- The agent can accept or reject. If reject, then the game ends. If accept, the agent choose a_1 for period 1.
- After realization of first period output, contract can be renegotiated.

Assuming there is no discounting, the principal creates a contract to maximize her utility function V:

$$\max_{w_i, w_{ij}, a_1, a_i} \sum_{i=1}^{n} p_i(a_1) \left(V(q_i^1 - w_i) + \sum_{j=1}^{n} p_j(a_i) V(q_i^2 - w_{ij}) \right), \tag{26}$$

subject to the **IC** incentive compatibility constraint, which is saying a_1, a_i needs to be the action that maximize the agent's utility, and **IR** individual rationality, which is saying that the agent need to obtain utility from the program that's at least as big as the next best outside option. Explicitly, **IC** is of the following form, where a_1 is the action in period 1 and a_2 is action in period 2 contingent upon $q^{(1)}$, the

probability distribution in period 1.

(27)
$$a_1, a_2 \in \arg\max_{a_1', a_2'} \sum_{i=1}^{n} p_i(a_1') \left(u(w_i) - \psi(a_1') + \sum_{j=1}^{n} p_j(a_2') u(w_{ij}) - \psi(a_2') \right),$$

and intuitively, this equation says the agent would choose the action a_1 and a_2 such that they maximize the RHS quantity. The sum within the brackets on the RHS of equation (27) is the expected payout of second period for the agent. We could theoretically calculate such a payout for each action taken in the first period, and then the sum outside the bracket just says we need to take the expected payout in the first period, while knowing the expected payout in the second period for each first period choice we make. Hence, on a high level view, this equation is similar to backwards induction.

Also, the **IR** could be expressed as

$$\sum_{i=1}^{n} p_i(a_1') \left(u(w_i) - \psi(a_1') + \sum_{j=1}^{n} p_j(a_2') u(w_{ij}) - \psi(a_2') \right) \geq \bar{u}, \tag{28}$$

where $\bar{u}$ is the next-best outside option for the agent. Notice that this equation says the agent's expected utility (gained from entering the contract with the principal) should be at least the amount of his next best option.

Solving the problem, one could obtain

$$\frac{V'(x_i - w_i)}{u'(w_i)} = \sum_{j=1}^{n} p_j(a_i) \frac{V'(x_j - w_{ij})}{u'(w_{ij})}, \tag{29}$$

if the principal is risk neutral, we could reduce to:

$$\frac{1}{u'(w_i)} = \sum_{j=1}^{n} p_j(a_i) \frac{1}{u'(w_{ij})}. \tag{30}$$

As in Bolton and Dewatripont (2004), we see that intuitively, the optimal contract smoothes the agent's consumption so as to equate the inverse of his marginal utility in the first period with the expected (average) inverse marginal utility in the second period. Hence, one benefit of multi-period contracting for the agent is the opportunity to take advantage of the inter-temporal arbitrage, so as to smooth consumption over the periods, and also choose a high level of effort (action) to hedge against potential low output in one period.

Remark in Bolton and Dewatripont (2004): in general, the optimal multi-period contract will be very complex. This complexity is contrasted with the simplicity of long-term incentive contracts observed in reality. Long term employment contracts observed in practice do not appear to fine-tune the agent's intertemporal consumption to the extent predicted by the model.

We are interested in record a rationale for linear contracts in T periods Stole (1999), which is in turn based on Holmstrom and Milgrom (1987). We consider an agent who choose a probability $p = (p_0, \ldots, p_t)$ for each period after observing all the history of outputs up until that time. The cost of effort is denoted by $c(p_\tau)$ for τ denoting running index for the periods. Let superscripts denote histories of variables, $X^t = \{x_0, \ldots, x_t\}$. The agent gets paid wage $w(X^t)$ at the end of the period, and has a cost of effort from each period: $\sum_{\tau=0}^{t} c(p_\tau)$. Thus, the utility of the agent is

$$U(w, p_1, \ldots, p_t) = -\exp\left(-r(w - \sum_{\tau=0}^{t} c(p_\tau)\right),$$

where exp denotes the exponential function, and we have assumed utility function U to take this form to simplify analysis (wealth effects). The take-away is that the optimal contract $p^* = p_t(X^{t-1})$ is **linear in accounts**, specifically,

$$w(X^T) = \sum_{t=1}^{T} w(x_t, p^*) = \sum_{i=0}^{N} w(x_i, p^*) \cdot A_i^T.$$

This is generally not linear in profits. If there are only two accounts, such as success or failure, then wages are linear in "profits", (i.e. successes).

Here we reproduce the same program but omitting the Stochastic bits in Holmstrom and Milgrom (1987). Consider a T-period model, in each period $t \in [T]$, the agent picks $p^t \in P$, incurring a cost that depends on the period $c(p^t)$. We denote the outcome x^t and the profit level π^t. We call $X^t = (x^1, \cdots, x^t)$ the history of the outcome process up to time t. Agent can observe X^{t-1} before deciding p^t. Thus, a strategy for the agent is a process $\{p^t(X^{t-1})\}$.

The principal pays the agent at the end of the last period based on the entire realized path X^T of the outcome process. The incentive

scheme is denoted $s(X^T)$. The agent's final wealth y is

$$y = s(X^T) - \sum_{t=1}^{T} c(p^t).$$

The agent has utility function $u(y) = -\exp(-ry)$. The principal has final wealth

$$\sum_{t=1}^{T} \pi^t - s(X^T),$$

and the principal has utility function $v(y) = -\exp(-Ry)$. The principal's problem is to choose a contract that maximizes her end of period utility, subject to the **IC** and **IR** for the agent. Hence, the principal maximizes

$$\max_{\{p^t\},s} \mathbb{E}\left(v\left(\sum_{t=1}^{T} \pi^t - s(X^T)\right)\right),$$

subject to the **IR** constraint

$$\mathbb{E}\left(u\left(s(X^T) - \sum_{t=1}^{T} c\left(p^t(X^{t-1})\right)\right)\right) \geq u(0),$$

where we have normalized the minimum certain equivalent to 0. And subject to the **IC**:

(31) $\{p^t\}$ maximize the left hand side of the equation above.

The expectations are taken with respect to the distribution over states induced by the agent's strategy $\{p^t\}$. The program is analyzed using dynamic programming.

2.3.1. *Remarks.* We see in the later sections that the explicit solution to a special case of Sannikov's model does not give predictions consistent with the rising wage profile. One explanation is that assuming infinite horizon has distorted the model's predictive power, since real world contracts are finite horizon, and agent's employment horizon is also finite. Some other elements present in discrete-time, multi-period models (and absent in our special case of Sannikov's model) are outside savings, richer set of actions by performing inter-temporal arbitrage, enhanced monitoring due to writing multiple contracts, etc. Those elements could be explored in a trackable way under the discrete time multi-period setting, and this is one of the advantages of this framework. For example, it is desirable to add savings to the model. Through out our discussion so far, we assumed that the agent has no outside savings options. Hence, in the model, the agent can only achieve saving

by signing a contract that have a payment scheme that fits the agent's desired consumption plans. Without access to credit and saving, the agent relies on the principal to allocate money over time, which would limit the agent's set of possible effort levels and actions. Hence, if we introduce savings and credit, then the agent would have a much richer set of actions, and would thus be able to pursue some different combinations of effort levels and achieve different contracting outcomes.

2.4. **Moral hazard problem continuous time, first best, one period.** We borrow freely from Stole (1999) lecture notes. Here we give the benchmark model with full information, which is first best. We assume the Agent's action (effort) is observable and verifiable. The principal makes a take-it-or-leave-it offer to an agent with next best utility of $\underline{U}$. If the agent accepts the contract, then he chooses an action $a \in \mathcal{A}$, which will lead to an outcome, $x \in \mathcal{X}$. the outcome x is measured as monetary profits, and we assume x is observable and varifiable. The principal writes a contract $w(x)$, which pays the agent wage $w(x)$ for outcome x. The agent takes an action a that gives a monetary disutility of $\psi(a)$, since we assume the agent dislike putting in action (effort), and the agent gets utility $u(\cdot)$ from wage $w(x)$. Assuming separability of utility in wage and cost of effort, the net utility of the agent is

$$U(w(x), a) \equiv u(w(x)) - \psi(a), \tag{32}$$

where several differentiability condition holds.

Let $\mathcal{X} = [x_1, x_2] \subset \mathbb{R}$ be the support of the probability density function $f(x, a)$, and let $F(x, a)$ be the cumulative distribution function. We assumes functions are infinitely differentiable C^∞, and $F_a(x, a) < 0$. Since this is first best, the principal chooses the agent's action a and contract $w(x)$ to satisfy

$$\max_{w(\cdot),a} \int_{\mathbb{R}} V(x - w(x)) f(x, a) dx,$$

where $V(\cdot)$ is principal's utility function, and $x - w(x)$ could be think of as the net monetary gain, which is monetary output minus wage paid to the agent. The **IR** (individual rationality) constraint/participation constraint is

$$\int_{\mathbb{R}} u(w(x)) f(x, a) dx - \psi(a) \geq \underline{U}, \tag{33}$$

and form the Lagrangian, obtain the first order condition

$$\frac{V'(x - w(x))}{u'(w(x))} = \lambda, \forall\ x \in \mathcal{X}, \tag{34}$$

which is known as the Borch rule: the ratios of marginal utilities of income are equated across states under an optimal insurance contract. It holds for every x in the support of the p.d.f.

2.5. **Moral hazard problem continuous time, second best, one period.** Now we update the previous model where the agent's action a cannot be observed and contracted upon, and the agent is risk averse, meaning $u'' < 0$. So the Principle faces an additional constraint, which is **IC**, incentive compatibility constraint for effort:

$$a \in \arg\max_{a' \in \mathcal{A}} \int_{\mathbb{R}} u(w(x)) f(x, a') dx - \psi(a'), \tag{35}$$

maximizing pointwise of the Lagrangian, we obtain the modified Borch rule:

$$\frac{V'(x - w(x))}{u'(w(x))} = \lambda + \mu \frac{f_a(x, a)}{f(x, a)}, \ \forall \ x \in \mathcal{X}, \tag{36}$$

where μ is the multiplier coming from **IC**. The modified Borch rule says, the marginal rates of substitution may vary if $\mu > 0$ to take into account the incentives effort of $w(x)$. Hence, risk-sharing will generally be inefficient.

2.6. **Moral hazard problem in stochastic time, second best.** We assume that the principal does not observe the agent's action A_t, and use the observations of outcome X_t to give the agent incentives to work hard. The main reference is Sannikov (2008), which is beyond my scope... The agent choose action (effort) A_t at every time $t \geq 0$, so the agent is choosing a stochastic process $\{A_t\}_{t \geq 0}$. Principal chooses a contract (which gives a flow of consumption for the agent) $\{C\}_{t \geq 0}$ based on output X_t, and the agent has action (effort) $\{A\}_{t \geq 0}$, which are all stochastic processes. We assume we can pick a suitable filtration of σ-algebra and all stochastic processes are adapted to this filtration. output X_t is also a stochastic process, assumed to be publically observable. Z_t denotes a standard Brownian motion that drives the output process. Output, agent's effort, and the Brownian motion Z_t evolving according to

$$dX_t = A_t dt + \sigma dZ_t, \tag{37}$$

σ is a constant. Cost of effort for the agent is denoted as $h(A_t)$, Where h maps the action of the agent A_t to a corresponding real number $\in \mathbb{R}$. Both the principal and the agent discount the flow of profit and utility

at rate r. The agent's average expected utility is given by

$$\mathbb{E}\left(r\int_0^\infty e^{-rt}\left(u(C_t)-h(A_t)\right)dt\right), \tag{38}$$

and the principal's problem is maximize her average profit given by:

$$\mathbb{E}\left(r\int_0^\infty e^{-rt}\left(A_t-C_t\right)dt\right), \tag{39}$$

subject to IR constraint of the agent:

$$\mathbb{E}\left(r\int_0^\infty e^{-rt}\left(u(C_t)-h(A_t)\right)dt\right)\geq \underline{U}, \tag{40}$$

and the IC constraint. In this setting, we say agent's effort A_t is incentive compatible with respect to the contract C_t if A_t maximizes the agent's total expected utility.

3. A special case of the continuous time model from Sannikov

3.1. **Motivation for Sannikov's model.** Assume the principal is risk neutral with constant utility function $u(x) = x$, and the agent is risk averse with unobservable action. We assume the agent's utility is separable in consumption and cost of action. Also, there is no saving in this model, so the agent cannot insure himself by saving. Also, the principal can only specify the contract based on output and time, but not on the action of the agent. However, in this deterministic case, output X is a deterministic function of action A, so the principal can compute the agent's action base on the observable output. Hence, effectively, she is able to contract upon agent's action directly, so she can simply write a contract that pays the agent based on how much action he put forth and thus solves the optimization problem. Hence, the question I would like to explore is what if the agent is not fully rational? The model presented in this section defers from the traditional models in the literature in that the agent is assumed to be "behavioral" instead of fully-rational. To be specific, instead of choosing action to maximize his life time utility level, the agent chooses his action according to a rule-of-thumb that depends mostly on prior beliefs and personal judgements of the agent.

3.1.1. *Preliminary set up of the deterministic case.* Let us start by presenting a fully rational, deterministic model of the principal-agent problem for motivation. Our main reference for how to approach this type of problems is Stokey (2008). Assume we have the following stationary law of motion, which is assumed to be deterministic and linear (in the action of the agent),

$$X(t) = X_0 + \mu A(t), \ \forall\, t \geq 0. \tag{41}$$

So the principal try to solve the following problem:

$$V(X,t) \equiv \max_{\omega(X,t)} \int_{t=0}^{t=\infty} e^{-rt}\left(X - \omega(X,t)\right) dt \tag{42}$$

$$\text{subject to } \int_{t=0}^{t=\infty} e^{-rt}\left(u(\omega) - h(A)\right) dt \geq K_0 \text{ and that} \tag{43}$$

$$A(t) \text{ maximizes } \int_{t=0}^{t=\infty} e^{-rt}\left(u(\omega) - h(A)\right) dt, \tag{44}$$

where

$X \equiv X(t)$ denotes the outcome, which is a function of time.

$A \equiv A(t)$ denotes the action of the agent, which is a function of time.

$\omega \equiv \omega(X,t)$ denotes the contract (ωage paid to agent), which is a function of X and t.

$X_0,\ K_0 \in \mathbb{R}$ is a constant.

Equation 43 is the IR (individual rationality) constraint, and it is saying that the agent will only enter the contract if he gets equal or more utility than his next best outside option K_0. Equation 44 is the IC (incentive compatible) constraint, and it is saying that the agent will choose the action $A(t)$ that maximizes his life-time utility.

Recall the (heuristic) Bellman's equation where Δt is a small time interval:

$$V(X,t) = (X - \omega(X,t))\Delta t + \frac{1}{1 + r\Delta t}\left(V(X,t) + \frac{\partial V(X,t)}{\partial t}\Delta t\right) \tag{45}$$

$$rV(X,t) = X - \omega(X,t) + \frac{\partial V(X,t)}{\partial t} \tag{46}$$

(47)

where we rearranged and canceled terms and let $\Delta t \to 0$ when we go from the first equation to the second equation. Equation 46 is **principal's Bellman equation**. Intuitively, during a small time interval, the right hand side is the "amount of dividend" generated from V, and

it equals the left hand side, which is the value generated (during this small time interval) plus any change in the value of V.

3.1.2. *Two intertwined Bellman equations.* For IR constraint, normalize to zero, so this constraint basically goes away, imagine an agent who is desperate, he will end up on the streets unless he could get this contract with the principal.

On the other hand, IC constraint is stating that the agent is also solving a maximization problem, which could be framed as a optimal stopping problem, in the sense that after entering the contract, the agent will choose to stop putting in effort once his continuation value dips below his next-best outside option K_0.

Let $W(t) \equiv W(\omega, t)$ denote the agent's continuation value. We sometimes suppress the ω as an argument since the wage ω ultimately depend on time, and we are in a deterministic setting. We have

$$W(t) \equiv W(\omega, t) \equiv \int_{s=t}^{s=\infty} e^{-r(s-t)} \left(u(\omega) - h(A)\right) ds. \tag{48}$$

Thus, **the agent has the following Bellman equation**:

$$W(t) = \left(u(\omega) - h(A)\right) \Delta t + \frac{1}{1 + r\Delta t} \left(W(t) + \frac{\partial W(t)}{\partial t} \Delta t \right) \tag{49}$$

$$rW(t) = u(\omega) - h(A) + \frac{\partial W(t)}{\partial t}. \tag{50}$$

Here, the interpretation of the Bellman equation is essentially the same as that of the principal. Now we would like to solve the following ODE:

$$\frac{\partial W(t)}{\partial t} = rW(t) - u(\omega) + h(A) \tag{51}$$

$$\text{with boundary condition } W(t) \geq K_0 \ \forall t. \tag{52}$$

The boundary condition comes from IR. Intuitively, if at any point in time, the agent's continuation value $W(t)$ dips below K_0, the agent could quit his job and pursue his next-best outside option that gives him K_0; alternatively, the agent could choose to put in zero effort and still enjoy the utility flow generated from ω, and wait till the principal to find out and fire the agent.

To solve equation 51, we need to know $\omega(X, t)$, which means the agent needs to know how the principal is going to decide the contract ω for all future time t, but $\omega(X, t)$ in turn depends on $X(t)$, which is driven by the agent's action. Hence, this inter-dependency makes the

program very hard to solve in closed form, and we need to make further assumptions on the agent's behavior, and this motivates the approach of bringing in social preferences.

Continue with the deterministic model, recall that output $X(t)$ is linear in the agent's action $A(t)$, we could make the following substitution in the Bellman equation 46 for the principal

$$\frac{\partial V}{\partial t} = rV - X + \omega \tag{53}$$

$$\frac{\partial V}{\partial t} = rV - X_0 - \mu A + \omega. \tag{54}$$

Hence, together with the Bellman equation for the agent, we have a system of Bellman equations:

$$\begin{cases} \frac{\partial V}{\partial t} & = rV - X_0 - \mu A + \omega. \\ \frac{\partial W}{\partial t} & = rW - u(\omega) + h(A) \end{cases} \tag{55}$$

Now, since this optimization problem is deterministic, and the principal knows the agent's action through the law of motion: $A(t) = \frac{1}{\mu}\left(X(t) - X_0\right)$. Hence, the principal could specify a contract that compensates the agent based on his effort $A(t)$. Without knowing the functional forms of $u(\cdot)$ and $h(\cdot)$, there is no obvious reason that the principal could align the incentives represented by the two equations. Hence, this approach fails, and using agent's continuation value as the state variable, as in Sannikov (2008), is a viable approach that circumvents this dilemma.

3.2. **Continuous time benchmark model solution in closed form.** Everything in this section is based on Sannikov (2008). We have the continuation value of the agent W_t, which is a stochastic process depending on C_t and A_t. It is of the form

$$W_t \equiv \mathbb{E}\left[\int_t^\infty e^{-r(s-t)}\left(u(C_s) - h(A_s)\right) ds \mid \mathcal{F}_t\right] \tag{56}$$

We are interested in the following functional form where we have CRRA utility and quadratic cost of effort function:

$$u(x) = \frac{x^{1-\gamma} - 1}{1-\gamma} \tag{57}$$

$$h(x) = x^2 \tag{58}$$

So $u(x)$ is concave, corresponds to Agent being risk averse. $h(x)$ is convex, as in Sannikov, and the Economic intuition is that working

more when we are already working a lot would be more costly and exhausting. Also, use Jensen inequality on concave, we have

$$\mathbb{E}[U(x)] < U(\mathbb{E}[x]) \tag{59}$$

gives risk averse.

3.2.1. *Preliminary set up of the Sannikov model.* Consider the dynamic principal-agent problem in continuous time. A standard Brownian motion Z_t drives the output process. The total output X_t produced up to time t is driven by the following process:

$$dX_t = A_t dt + \sigma dZ_t, \tag{60}$$

where $\sigma \in \mathbb{R}$ is a constant.

- The State at time t is (t, W_t)
- The Action at time t is recommended action and the contract (compensation, consumption for the agent) together as a 2-tuple: $(a(W_t), c(W_t))$
- The Reward per unit time for the principal is $(A_t - C_t)$, representing the value created by the agent A_t minus the compensation (contract) C_t paid to the agent.

The Return (profit for the principal) at time t is the accumulated discounted Reward:

$$r \int_0^\infty e^{-rt}(A_t - C_t)dt \tag{61}$$

Recall the SDE in Sannikov that drives the entire stochastic optimization problem:

$$dW_t = r(W_t - u(C_t) + h(A_t))dt + r\sigma Y_t dZ_t. \tag{62}$$

Notice that $Y_t \equiv Y_t(W_t)$ is a stochastic process representing the sensitivity of the agent's continuation value to output. It's existence is established using Martingale Representation theorem in Sannikov. Therefore, this equation captures the infinitesimal change in the agent's continuation value dW_t from time t to time $t + dt$.

Our goal is to determine the optimal contract $\{a(W_t), c(W_t)\}$ (which is a 2-tuple that specifies the agent's recommended action and compensation stream) that maximize the principal's profit:

$$\mathbb{E}\left[r \int_0^\infty e^{-rt}(A_t - C_t)dt\right] \tag{63}$$

and satisfy the Incentive Compatibility constraint, which says the agent's value function below should also be maximized.

$$\mathbb{E}\left[r\int_0^\infty e^{-rt}(u(C_t)-h(A_t))dt\right] \tag{64}$$

We could interpret the drift as the difference between future continuation value and the current value, which is $u-h$, utility minus cost of effort. Let $V^*\equiv V^*(t,W_t)$ denote the optimal value function of the principal. The HJB equation is of the following form:

$$r\cdot V^*(t,W_t)dt=\max_{c_t}\left\{\mathbb{E}[dV^*(t,W_t)]+(A_t-C_t)dt\right\} \tag{65}$$

We write $V^*\equiv V^*(t,W_t)$ for simplicity. We apply Ito's Lemma on dV^*:

$$dV^*=\left(\frac{\partial V^*}{\partial t}+r(W_t-u(C_t)+h(A_t))\frac{\partial V^*}{\partial W_t}+\frac{r^2\sigma^2Y_t^2}{2}\frac{\partial^2V^*}{\partial W_t^2}\right)\cdot dt \tag{66}$$

$$+r\sigma Y_t\frac{\partial V^*}{\partial W_t}dZ_t \tag{67}$$

Substitute equation 66 into equation 65, the volatility term (67) with dZ_t vanishes after we take the expectation $\mathbb{E}(\cdot)$ because it's a martingale. We further drop the dt term from both sides to obtain the following HJB Equation in partial differential form.

$$rV^*=\max_{c_t}\left(\mathbb{E}\left[\frac{\partial V^*}{\partial t}+r(W_t-u(C_t)+h(A_t))\frac{\partial V^*}{\partial W_t}+\frac{r^2\sigma^2Y_t^2}{2}\frac{\partial^2V^*}{\partial W_t^2}\right]+rA_t-rC_t\right). \tag{68}$$

Hence, we have recovered the HJB equation as in Sannikov (2008).

3.2.2. *CRRA utility benchmark solution.* Now, we proceed to obtain the benchmark case where the principal's value function is linear in agent's continuation value, when we assume the utility function is CRRA, and the cost of effort function is quadratic. This is saying that the principal's value (profit) increases when the agent's continuation value increases. Intuitively, this makes economic sense. When agent's wage is more back-loaded, the agent has long run incentive and a high continuation value, meaning the agent will get high reward if the agent do not shirk and work hard for the principle to the very end. Hence, the principal also enjoys a higher return when the agent has higher continuation value.

Proposition: Assuming the set up as discussed previously, we further assume the following functional form for agent's utility function

and cost of effort function:

$$u(C_t) \equiv \frac{C_t^{1-\gamma}}{1-\gamma} \tag{69}$$

$$h(A_t) \equiv A_t^2. \tag{70}$$

Also, we let $Y_t \equiv h'(A_t) \equiv 2A_t$ using the same reasoning as in Sannikov. If the above is satisfied, then the principal's optimal value function V^* is linear in agent's continuation value W_t, and is of the following form:

$$V^*(t, W_t) = \left(\frac{-1}{4\eta}\right)^{\frac{\gamma}{1+\gamma}} W_t, \tag{71}$$

where

$$\eta \equiv \frac{1}{1-\gamma}(-1)^{(1-\gamma)/\gamma} + (-1)^{1/\gamma}. \tag{72}$$

Proof. The proof is mainly computation, and is deferred to Appendix A. □

3.2.3. *Intuition behind the benchmark solution.* For the benchmark stochastic model, I need $Y_t \equiv h'(A_t)$, and the rationale could be attributed to the behavioral update, namely, how the agent will work as little as possible so that he does not expose himself too much to risk, relying on a rule-of-thumb.

Economic intuition for the CRRA benchmark closed form solution is as follows. The principal's value function is linear in agent's continuation value says that if the principal could give the agent a higher continuation value, then the principal in turn enjoys a proportionally higher value. Hence, the principal would choice a contract so that the agent's wage is back-loaded. Because of wealth effect, if the principal rewards the agent with a high wage in the near term periods, then it requires more wage to induce the same level of effort from the agent since the agent had become richer. Thus, if the wages are back loaded, then wealth effect is less prominent, and the agent has more reason to stick to the contract and continue to supply high effort, since the agent knows he will collect larger rewards in the future. Hence, this analysis implies that long term incentives such as opportunities of promotion and higher wage category for experienced workers are going to increase agent's continuation value, leading to an increase in principal's value function.

3.2.4. *Special case of CRRA utility: log utility.* We know that log utility is a special case of CRRA utility, and we expect the solution to be behaving in the same manner as the more general CRRA utility.

Proposition: Assuming the set up as discussed previously, we further assume the following functional form for agent's utility function and cost of effort function:

$$u(C_t) = \log C_t \tag{73}$$

$$h(A_t) = A_t^2. \tag{74}$$

Again, using $Y_t = h'(A_t) = 2A_t$, we have the following HJB differential equation. Also, we let $Y_t \equiv h'(A_t) \equiv 2A_t$ using the same reasoning as in Sannikov. If the above is satisfied, then the principal's optimal value function V^* is linear in agent's continuation value W_t, and is of the following form:

$$V^*(W_t) = K_1^r W_t \tag{75}$$

where K_1 is a constant that needs to satisfy the following equation:

$$0 = -K_1^r \pi i - K_1^r r \log K_1 - \frac{1}{4} K_1^{-r} + K_1^r. \tag{76}$$

Proof. The proof is mainly computation, and we defer it to the Appendix B. □

However, our result is not very strong, considering the fact that it does not conciliate with the rising wage profile theory. As workers gain job seniority by staying in the same roles, their wage would rise relative to other job alternatives, as in Topel (1991). As recorded in Appendix D, our result predicts that the wage/consumption profile would stay constant no matter how long a worker is on the same job, which is a contradiction with the prevalent views found in rising wage profile arguments.

4. Concluding Remarks

Loewenstein and Sicherman (1991) gives a nice overview of some literature pertaining to the rising wage profile argument, and in this article, the authors also discussed why workers might prefer increasing wage profiles, despite it having a lower discounted present-day value than a decreasing wage profile. Moreover, as in Lazear (1976), as workers age and gain human capital on the job, their wage will also increase accordingly. Hence, it is natural to expect that a worker's wage should be a increasing function with respect to time t. However, as shown in Appendix D, the optimal wage/contract (which is taken to be the

same as consumption) path C^* is constant for our closed form solution. Hence, our solution in the CRRA special case does not reconciliate well with the canonical economic theory. Intuitively, the reason behind this contradiction is that in this model, the Principal and the Agent come together at the start of time and sign a contract, which goes on forever, following a stochastic path. Although infinite horizon has brought us some analytical advantages by allowing us to do calculus, it does introduce elements that are not applicable to the real world. For example, unless forcing the agent to exit the contract or retire, the principal would be paying the agent an infinite amount of money by keeping the agent's continuation value high at all periods. Since labor contracts always have a finite time horizon in the real world, it is understandable that our explicit solution in the CRRA case of this stochastic model gives some distorted predictions. Hence, discrete time multi-period models in Section 2.3 might offer a more accurate view (and avoid using stochastic calculus) of what is going on in a contracting agreement between a Principal and an Agent over an extended period of time.

Appendix A. Computations for the CRRA utility benchmark solution in closed form

To obtain the CRRA utility benchmark case, we use the following specific functional form for agent's utility function and cost of effort function:

$$u(C_t) \equiv \frac{C_t^{1-\gamma}}{1-\gamma} \tag{77}$$

$$h(A_t) \equiv A_t^2. \tag{78}$$

Also, we let $Y_t \equiv h'(A_t) \equiv 2A_t$ using the same reasoning as in Sannikov. Hence, HJB equation is of the following specific form:

$$rV^* = \max_{C_t,\ A_t} \left(\mathbb{E}\left[\frac{\partial V^*}{\partial t} + r\left(W_t - \frac{C_t^{1-\gamma}}{1-\gamma} + A_t^2 \right) \frac{\partial V^*}{\partial W_t} + \frac{4r^2\sigma^2 A_t^2}{2} \frac{\partial^2 V^*}{\partial W_t^2} \right] + rA_t - rC_t \right) \tag{79}$$

The first order condition with respect to the right hand side gives:

$$r\frac{\partial V^*}{\partial W_t} C_t^{-\gamma} - r = 0 \tag{80}$$

$$r\frac{\partial V^*}{\partial W_t} 2A_t + 4r^2\sigma^2 A_t \frac{\partial^2 V^*}{\partial W_t^2} + r = 0, \tag{81}$$

and re-arranging gives

$$C_t^* = \left(-\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma} \tag{82}$$

$$A_t^* = -\left(2\frac{\partial V^*}{\partial W_t} + 4r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right)^{-1} \tag{83}$$

Now we substitute C_t^* and A_t^* back into our HJB equation to obtain the following Partial Differential Equation:

$$rV^* = \frac{\partial V^*}{\partial t} \tag{84}$$

$$+ r\left(W_t - \frac{1}{1-\gamma}\left(-\frac{\partial V^*}{\partial W_t}\right)^{(1-\gamma)/\gamma} + \left[2\frac{\partial V^*}{\partial W_t} + 4r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right]^{-2}\right)\frac{\partial V^*}{\partial W_t} \tag{85}$$

$$+ 2r^2\sigma^2\frac{\partial^2 V^*}{W_t^2}\left(2\frac{\partial V^*}{\partial W_t} + 4r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right)^{-2} \tag{86}$$

$$+ (-r)\left(2\frac{\partial V^*}{\partial W_t} + 4r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right)^{-1} \tag{87}$$

$$+ (-r)\left(-\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma}. \tag{88}$$

After some meticulous grouping of terms, we have

$$rV^* = \frac{\partial V^*}{\partial t} + rW_t\frac{\partial V^*}{\partial W_t} \tag{89}$$

$$- \frac{r}{1-\gamma}(-1)^{(1-\gamma)/\gamma}\left(\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma} \tag{90}$$

$$- \frac{r}{4}\cdot\left(\frac{\partial V^*}{\partial W_t} + 2r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right)^{-1} \tag{91}$$

$$- r\cdot\left(-\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma} \tag{92}$$

We use the following short hand:

$$\eta \equiv \frac{1}{1-\gamma}(-1)^{(1-\gamma)/\gamma} + (-1)^{1/\gamma}. \tag{93}$$

(Notice that for some values of γ, η could be of complex value instead of real value. Still, we could take values such as $\gamma = 0.5$ and in this case $\eta = -1$ is of real value.)

We collect terms to obtain:

(94)
$$rV^*(t, W_t) = \frac{\partial V^*}{\partial t} + rW_t \frac{\partial V^*}{\partial W_t} - r\eta \left(\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma} - \frac{r}{4} \cdot \left(\frac{\partial V^*}{\partial W_t} + 2r\sigma^2 \frac{\partial^2 V^*}{\partial W_t^2}\right)^{-1}$$

We make the following judicious (or simple-minded) guess of the value function:

$$V^* = f(t)^r W_t, \tag{95}$$

and we obtain the following partial derivatives:

$$\frac{\partial V^*}{\partial t} = rf(t)^{r-1} f'(t) W_t \tag{96}$$

$$\frac{\partial V^*}{\partial W_t} = f(t)^r \tag{97}$$

$$\frac{\partial^2 V^*}{\partial W_t^2} = 0. \tag{98}$$

Hence, we obtain the following simplified form:

$$rV^* = \frac{\partial V^*}{\partial t} + rW_t \frac{\partial V^*}{\partial W_t} - r\eta \left(\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma} - \frac{r}{4} \left(\frac{\partial V^*}{\partial W_t}\right)^{-1} \tag{99}$$

$$rf(t)^r W_t = rf(t)^{r-1} f'(t) W_t + rW_t f(t)^r - r\eta f(t)^{r/\gamma} - \frac{r}{4} f(t)^{-r} \tag{100}$$

$$0 = rf(t)^{r-1} f'(t) W_t - r\eta f(t)^{r/\gamma} - \frac{r}{4} f(t)^{-r} \tag{101}$$

$$0 = f(t)^{r-1} f'(t) W_t - \eta f(t)^{r/\gamma} - \frac{1}{4} f(t)^{-r} \tag{102}$$

Hence, we see that $f(t) \equiv K$ where K is a constant would satisfy this equation. Thus, we solve for K, and we have

$$0 = -\eta K^{r/\gamma} - \frac{1}{4} K^{-r} \tag{103}$$

$$\eta K^{r/\gamma} = -\frac{1}{4} K^{-r} \tag{104}$$

$$K^{\frac{r}{\gamma} + r} = -\frac{1}{4\eta} \tag{105}$$

$$K = \left(\frac{-1}{4\eta}\right)^{\frac{\gamma}{r + r\gamma}} \tag{106}$$

Hence, the principal's value function is

$$V^*(t, W_t) = f(t)^r W_t = \left(\frac{-1}{4\eta}\right)^{\frac{\gamma}{1+\gamma}} W_t \tag{107}$$

Hence, we have shown that the principal's value function is independent of time (still implicitly depend on t since W_t is a stochastic process) and linear in agent's continuation value W_t.

Appendix B. Computations for the Log utility benchmark solution in closed form

For the special case in CRRA utility when $\gamma = 1$, we have log utility, and we still assume the cost of effort function to be quadratic:

$$u(C_t) = \log C_t \tag{108}$$

$$h(A_t) = A_t^2. \tag{109}$$

Again, using $Y_t = h'(A_t) = 2A_t$, we have the following HJB differential equation.

$$rV^* = \max_{C_t,\ A_t} \left(\mathbb{E}\left[\frac{\partial V^*}{\partial t} + r\left(W_t - \log C_t + A_t^2\right) \frac{\partial V^*}{\partial W_t} + \frac{4r^2\sigma^2 A_t^2}{2} \frac{\partial^2 V^*}{\partial W_t^2} \right] + rA_t - rC_t \right) \tag{110}$$

Take FOC of RHS with respect to C_t and A_t respectively, we obtain:

$$C_t^* = -\frac{\partial V^*}{\partial W_t} \tag{111}$$

$$A_t^* = -\left(2\frac{\partial V^*}{\partial W_t} + 4r\sigma^2 \frac{\partial^2 V^*}{\partial W_t^2}\right)^{-1} \tag{112}$$

The condition on C_t^* could be interpreted as a form of the Borch rule. Similar to the general CRRA case, we substitute the optimal contract C_t and the recommended action A_t to the HJB equation to obtain:

(113)

$$rV^* = \frac{\partial V^*}{\partial t}$$

(114) $$+ r\left(W_t - \log\left(-\frac{\partial V^*}{\partial W_t}\right) + \left[2\frac{\partial V^*}{\partial W_t} + 4r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right]^{-2}\right)\frac{\partial V^*}{\partial W_t}$$

(115) $$+ 2r^2\sigma^2\frac{\partial^2 V^*}{W_t^2}\left(2\frac{\partial V^*}{\partial W_t} + 4r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right)^{-2}$$

(116) $$+ (-r)\left(2\frac{\partial V^*}{\partial W_t} + 4r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right)^{-1}$$

(117) $$+ r\frac{\partial V^*}{\partial W_t}.$$

After some meticulous grouping of terms, we have

(118) $$rV^* = \frac{\partial V^*}{\partial t} + rW_t\frac{\partial V^*}{\partial W_t}$$

(119) $$- r\log\left(-\frac{\partial V^*}{\partial W_t}\right)\frac{\partial V^*}{\partial W_t}$$

(120) $$- \frac{r}{4}\cdot\left(\frac{\partial V^*}{\partial W_t} + 2r\sigma^2\frac{\partial^2 V^*}{\partial W_t^2}\right)^{-1}$$

(121) $$+ r\frac{\partial V^*}{\partial W_t}.$$

We proceed with the guess $V^* \equiv f(t)^r W_t$ to obtain:

(122) $$rf(t)^r W_t = rf(t)^{r-1}f'(t)W_t + rW_t f(t)^r$$

(123) $$- r\log\left(-f(t)^r\right)f(t)^r$$

(124) $$- \frac{r}{4}\cdot f(t)^{-r}$$

(125) $$+ rf(t)^r$$

We collect together coefficient before W_t term and constant term, and let $\log(-1) = \pi i$, where i denotes $\sqrt{-1}$. Hence, we have the following system of differential equations:

(126) $$\begin{cases} f(t)^r &= f(t)^{r-1}f'(t) + f(t)^r \\ 0 &= -r\log\left(-f(t)^r\right)f(t)^r - \frac{r}{4}f(t)^{-r} + rf(t)^r \end{cases}$$

Write $f \equiv f(t)$, the second equation simplify to

$$0 = -\log\left(-f(t)^r\right)f(t)^r - \frac{1}{4}f(t)^{-r} + f(t)^r \tag{127}$$

$$0 = -f^r\pi i - f^r r\log f - \frac{1}{4}f^{-r} + f^r \tag{128}$$

From the first equation, $f \equiv K_1$ for some constant K_1 would do the job. Then K_1 need to further satisfy equation 128.

Appendix C. A possible generalization of the CRRA utility benchmark solution

Here is the HJB equation reproduced for easy reference.

$$rV^* = \max_{c_t}\left(\mathbb{E}\left[\frac{\partial V^*}{\partial t} + r(W_t - u(C_t) + h(A_t))\frac{\partial V^*}{\partial W_t} + \frac{r^2\sigma^2 Y_t^2}{2}\frac{\partial^2 V^*}{\partial W_t^2}\right] + rA_t - rC_t\right) \tag{129}$$

If we are not allowed to infer anything on the stochastic process Y_t and proceed in greater generality, the following is what we would find.

We take derivative of RHS with respect to C_t and A_t and set equal to zero to obtain the following First Order Condition:

$$-u'(C_t)r\frac{\partial V^*}{\partial W_t} - r = 0 \tag{130}$$

$$h'(A_t)r\frac{\partial V^*}{\partial W_t} + r = 0 \tag{131}$$

And recall our SDE (which is from Proposition 1 in the Appendix of Sannikov):

$$dW_t = r(W_t - u(C_t) + h(A_t))dt + r\sigma Y_t dZ_t \tag{132}$$

Now we use specific functional form, and we can take $\gamma = 2$

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma} \tag{133}$$

$$h(x) = x^\gamma \tag{134}$$

We have

$$C_t^* = \left(-\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma} \tag{135}$$

$$A_t^* = \left(-\frac{1}{\gamma}\right)^{1/(\gamma-1)}\left(\frac{\partial V^*}{\partial W_t}\right)^{-1/(\gamma-1)} \tag{136}$$

We substitute the optimal value C_t^* and A_t^* back into the HJB equation to get

$$rV^* = \frac{\partial V^*}{\partial t} + r\left[W_t\frac{\partial V^*}{\partial W_t} + \frac{1}{1-\gamma}\left(-\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma} + \left(-\frac{1}{\gamma}\right)^{\gamma/(\gamma-1)}\left(\frac{\partial V^*}{\partial W_t}\right)^{-1/(\gamma-1)}\right] \tag{137}$$

$$+ r\left(-\frac{1}{\gamma}\right)^{1/(\gamma-1)}\left(\frac{\partial V^*}{\partial W_t}\right)^{-1/(\gamma-1)} - r\left(-\frac{\partial V^*}{\partial W_t}\right)^{1/\gamma} \tag{138}$$

$$+ \frac{r^2\sigma^2 Y_t^2}{2}\frac{\partial^2 V^*}{\partial W_t^2} \tag{139}$$

Now, it's almost impossible to guess the functional form of V^* since we do not know the stochastic process Y_t, hence we need further assumptions on Y_t like those presented in Sannikov. In order to obtain explicit solutions to the model, we would benefit from having an explicit functional form for the process Y_t. Hence, we see that the approach taken in Sannikov is quite necessary.

On the other hand, if we allow different assumptions regarding social preference, we could potentially obtain vastly different dynamics on the stochastic process Y_t, which would in turn give us a very different SDE that drives the optimization problem. Hence, the assumptions on functional forms of Y_t (or the lack there of) are the keys in determine the optimal contract and the principal's optimal value function, so a possible future direction is to using different functional forms for Y_t by tweaking the assumptions about the social preference of the principal and the agent.

Appendix D. Optimal consumption path and recommended action path for CRRA utility

After obtaining the closed form of the principal's value function V^*, we could substitute it back into the optimal consumption and optimal action functions to get the exact form. However, it turns out that they

are both constants, as follows:

$$C_t^* \equiv \left[-\left(\frac{-1}{4\eta} \right)^{\gamma/(1+\gamma)} \right]^{1/\gamma}, \tag{140}$$

$$A_t^* \equiv -\left(2 \left(\frac{-1}{4\eta} \right)^{\gamma/(1+\gamma)} \right)^{-1}. \tag{141}$$

Hence, this implies that over time, the solution predicts that the agent maintains the same level of consumption and same level of action (effort) ever since the contract starts. This contradicts the widely accepted rising wage profile theory, and worker's preference for rising wage profiles, as documented in Loewenstein and Sicherman (1991). Hence, our explicit solution in the CRRA utility special case does not give much explanatory power.

Similarly, optimal consumption C_t^* and optimal recommended action A_t^* are both constants under the special case of log utility. Hence, comparative statics such as high γ versus low γ simply adjusts the level of consumption and action to a different constant. Again, this benchmark model does not have much predictive power over optimal consumption path C^* and recommended action path A^*.

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